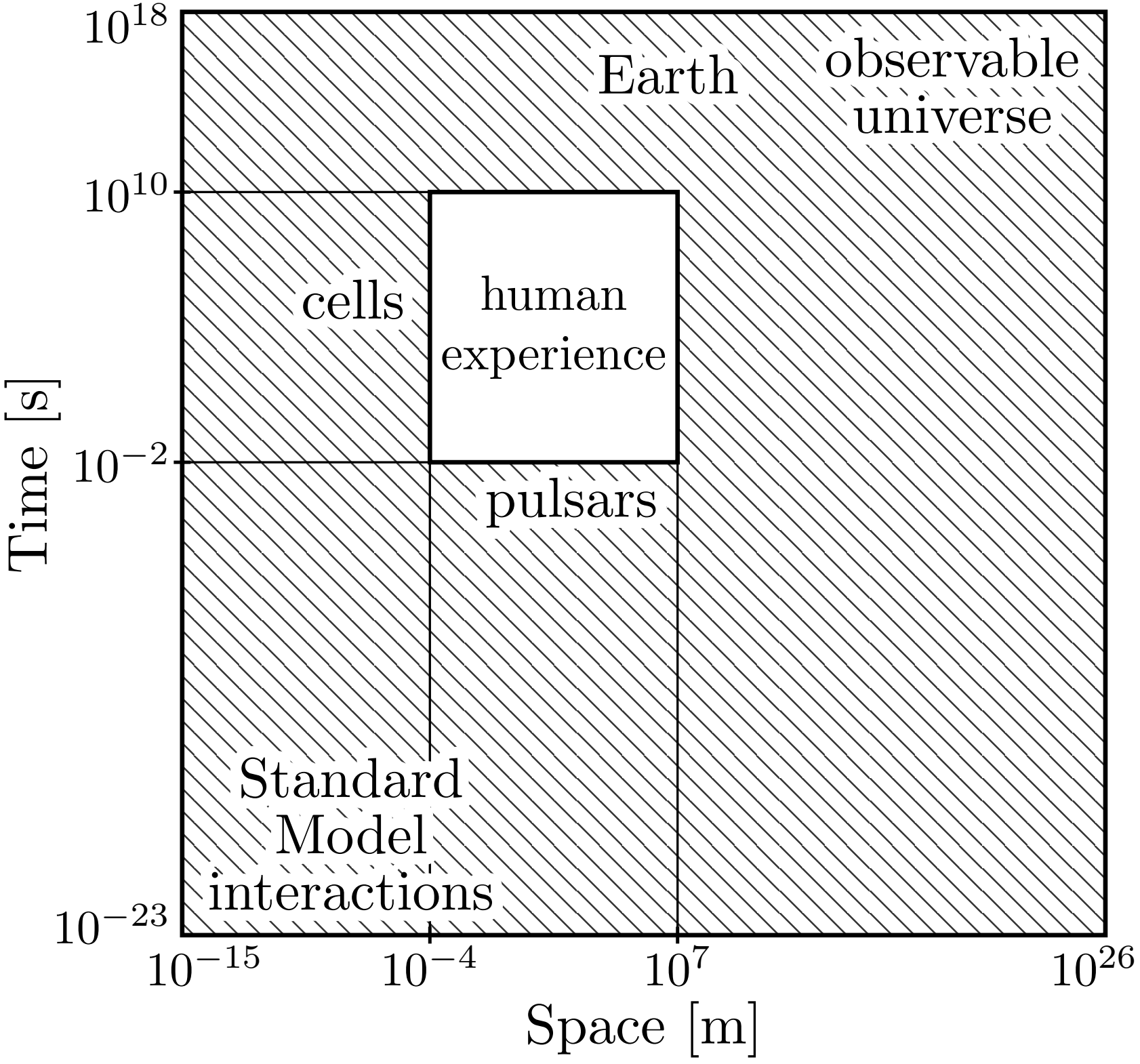
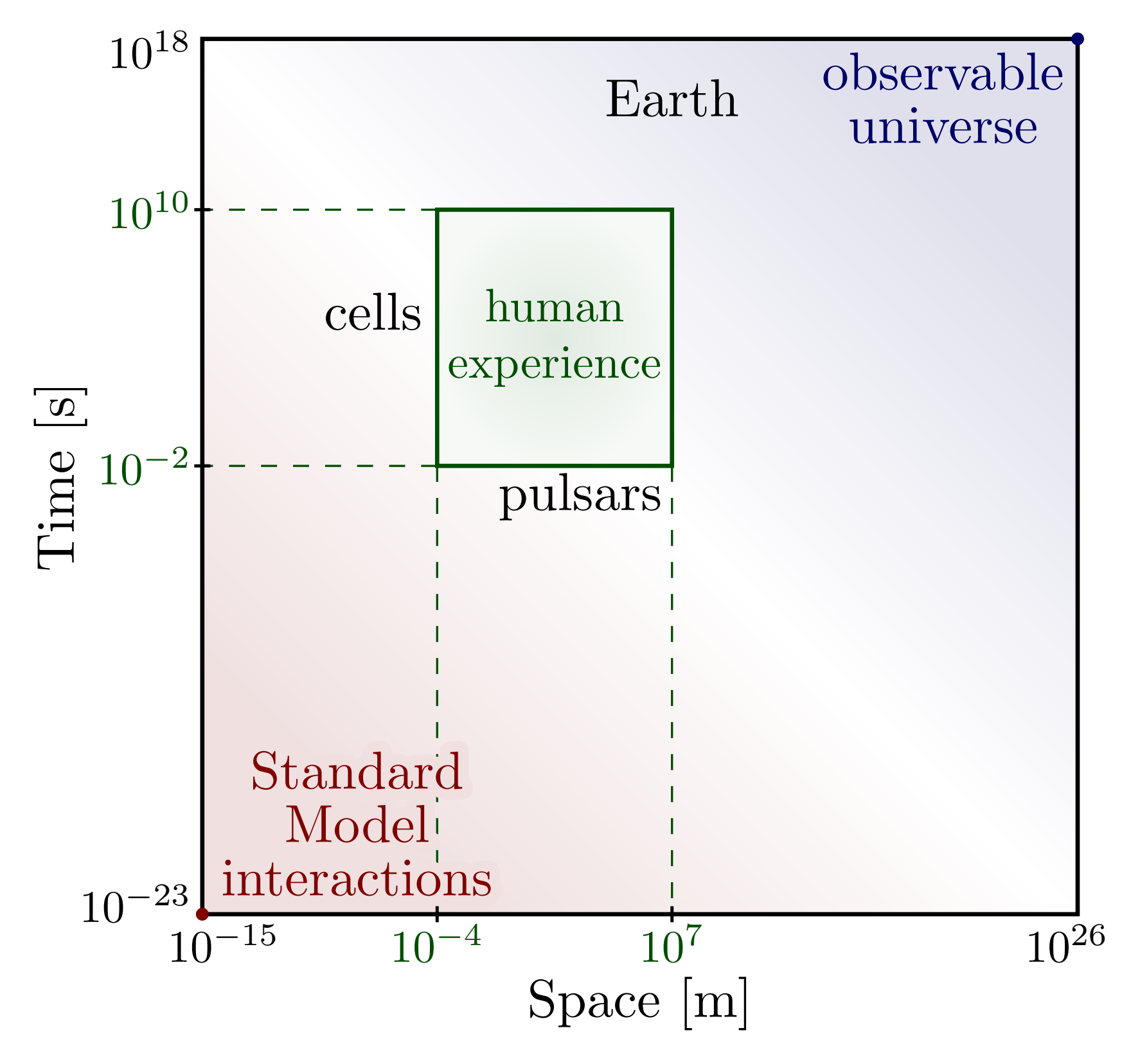
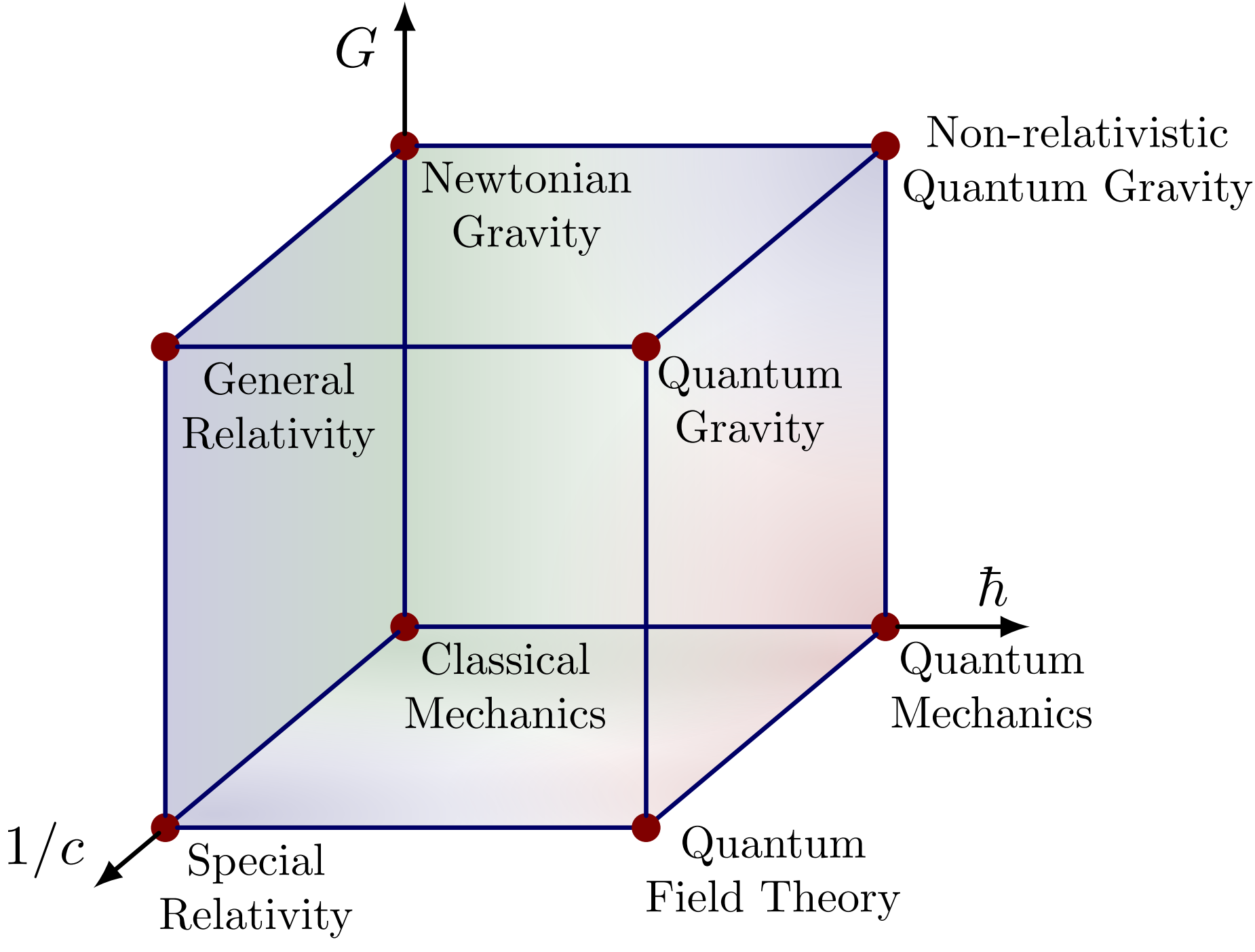
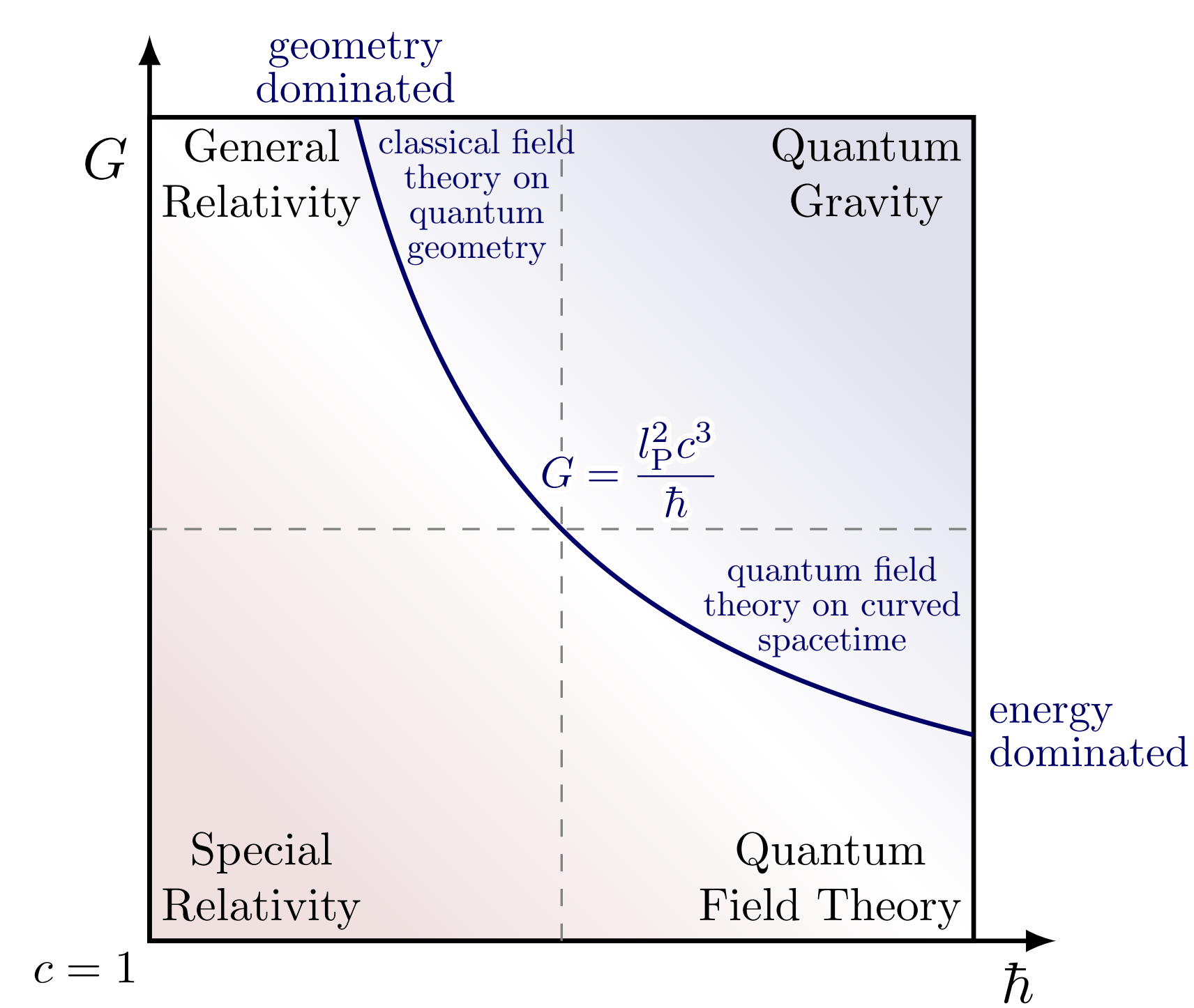
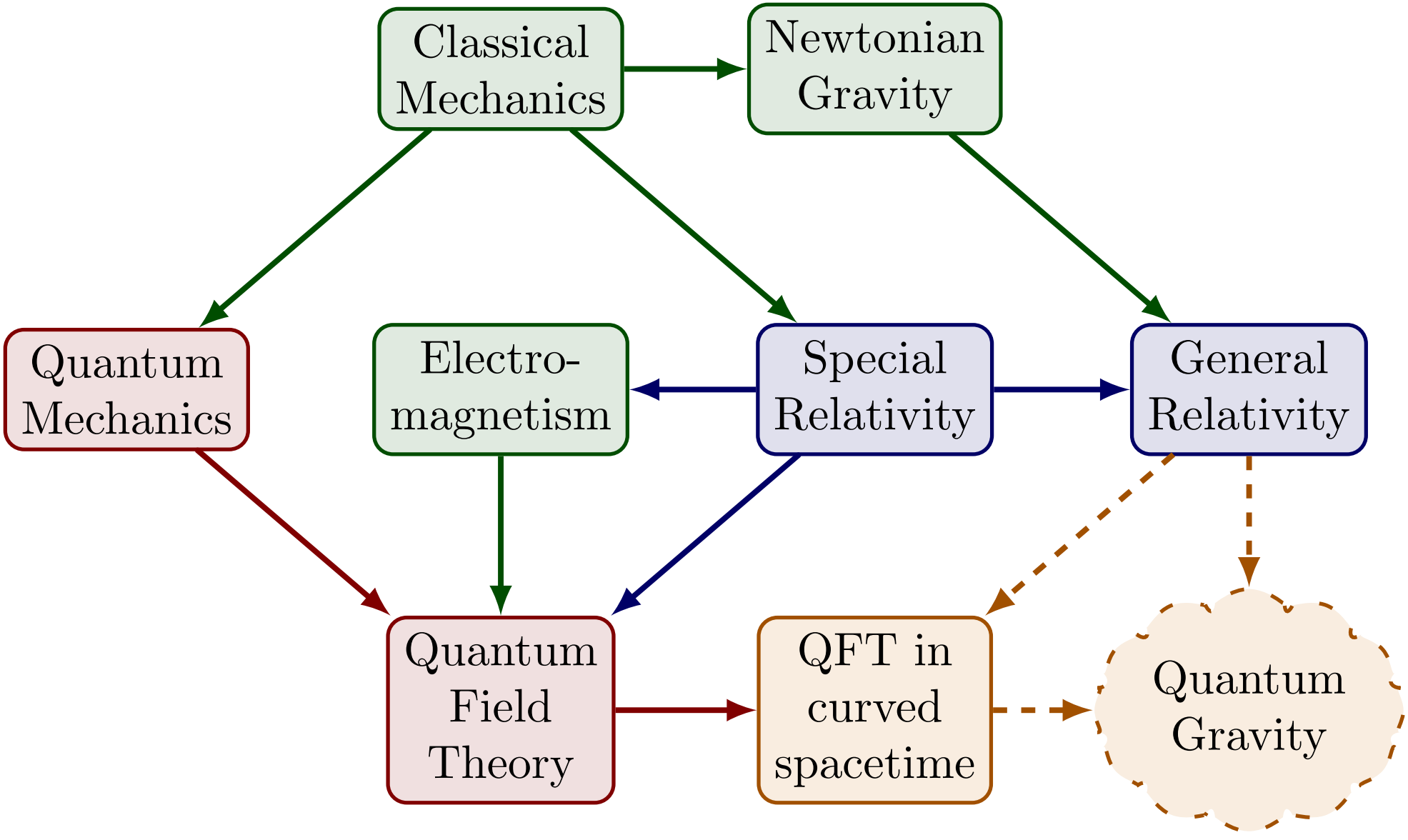
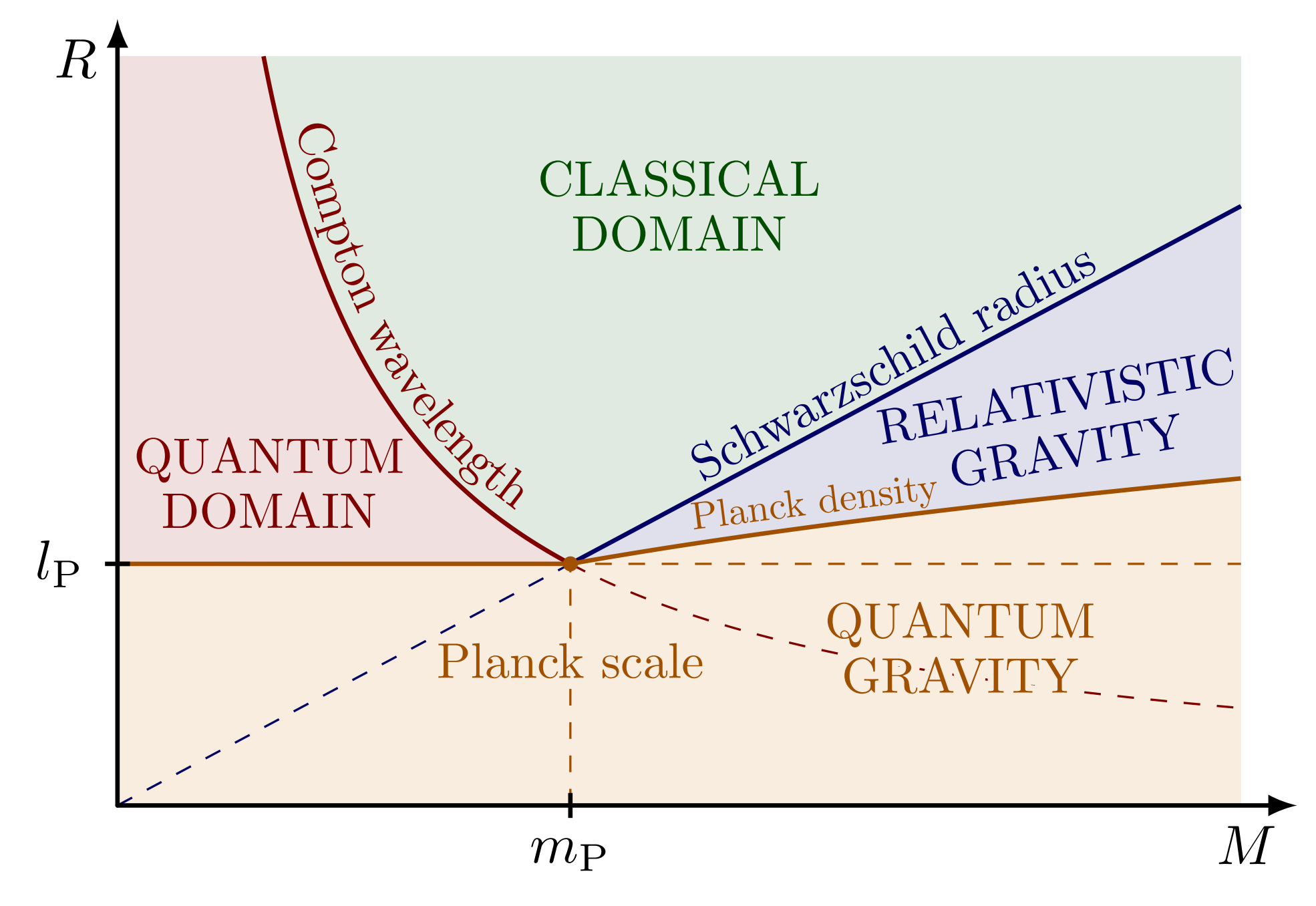

10^{18}
10^{10}
10^{-2}
10^{-23}
Time [s]
Earth
observable
universe
cells
human
experience
pulsars
Standard
Model
interactions
10^{-15}
10^{-4}
10^{7}
10^{26}
Space [m]

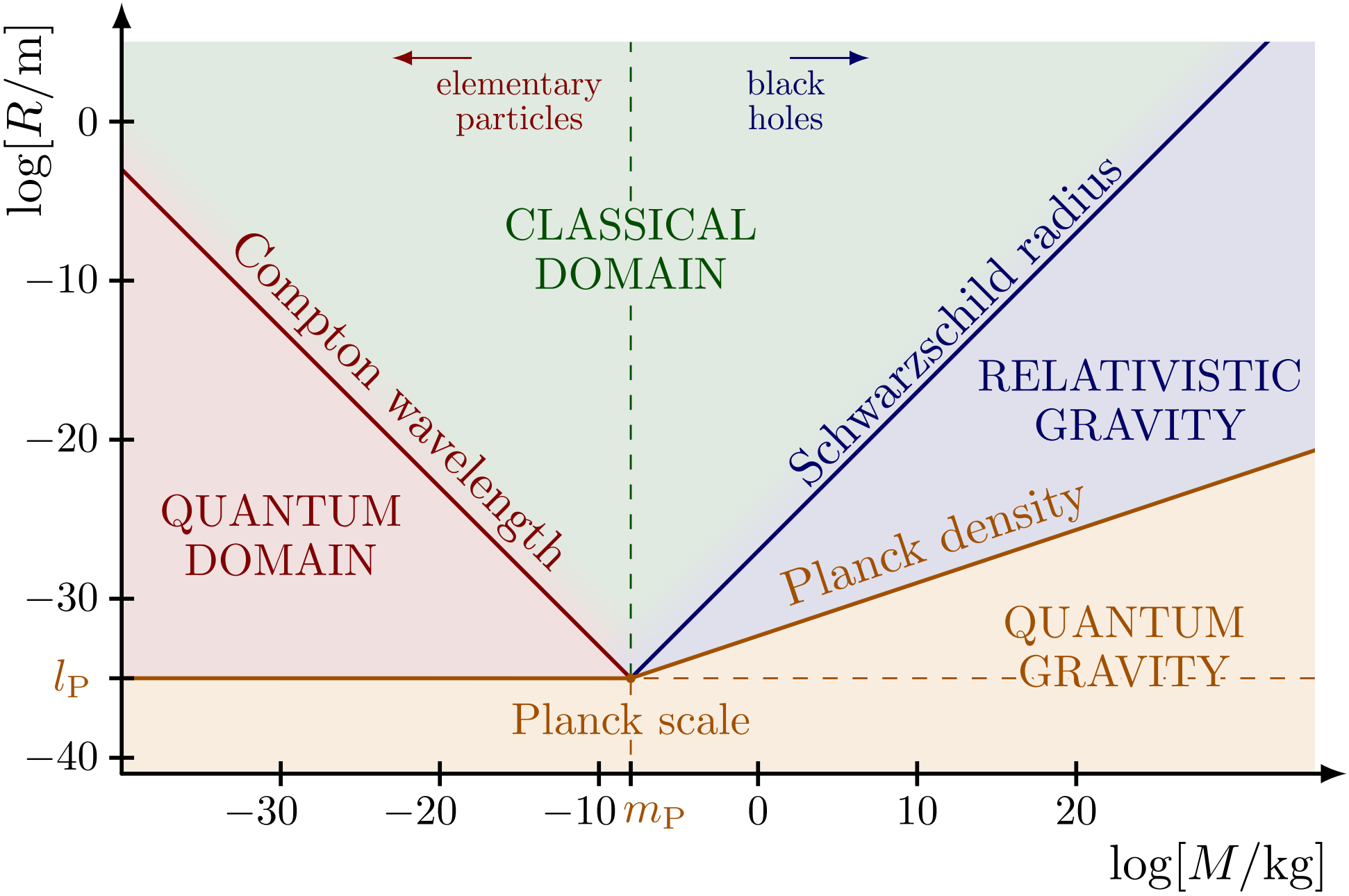

10^{18}
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interactions
10^{-15}
10^{-4}
10^{7}
10^{26}
Space [m]


G
Newtonian Gravity
Non-relativistic Quantum Gravity
General Relativity
Quantum Gravity
$\hbar$
Classical Mechanics
Quantum Mechanics
$1/c$
Special Relativity
Quantum Field Theory


geometry
dominated
G
General
Relativity
classical field
theory on
quantum
geometry
Quantum
Gravity
$G = \frac{l_{\mathrm{P}}^2 c^3}{\hbar}$
quantum field
theory on curved
spacetime
energy
dominated
Special
Relativity
Quantum
Field Theory
$c = 1$
$\hbar$


Classical Mechanics
Newtonian Gravity
Quantum Mechanics
Electro-magnetism
Special Relativity
General Relativity
Quantum Field Theory
QFT in curved spacetime
Quantum Gravity


R
Compton wavelength
CLASSICAL
DOMAIN
Schwarzschild radius
RELATIVISTIC
GRAVITY
Planck density
QUANTUM
DOMAIN
l_{P}
Planck scale
QUANTUM
GRAVITY
m_{P}
M


log[R/m]
0
−10
−20
−30
l_{P}
−40
elementary particles
black holes
CLASSICAL DOMAIN
Compton wavelength
Schwarzschild radius
RELATIVISTIC GRAVITY
QUANTUM DOMAIN
Planck density
QUANTUM GRAVITY
Planck scale
−30
−20
−10
m_{P}
0
10
20
log[M/kg]